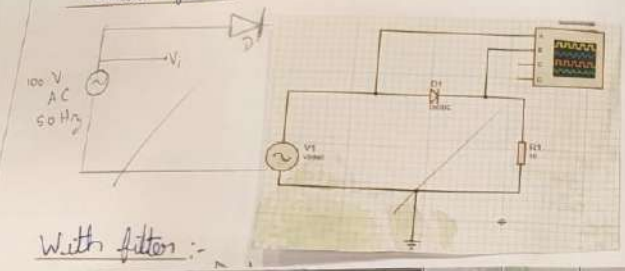
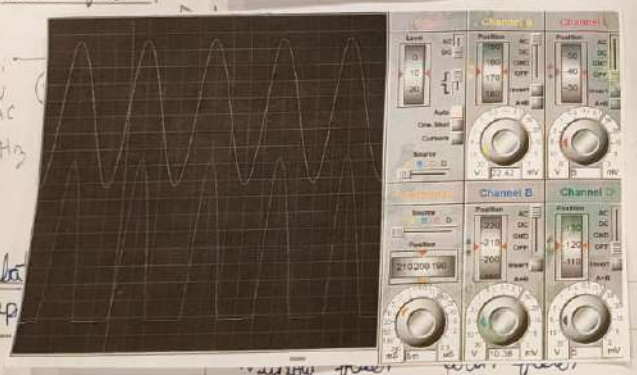
Circuit Diagram

Half rectifier :-

Without filter:



With filter :-



Tabulat

Inp

$V_M(t)$	t (MS)	V_H (V)	t (MS)	V_M (V)	t (MS)
100 V	20 MS	100 V	100 MS	100V, 85 V	5 ms, 25 ms

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HALF WAVE AND FULL WAVE RECTIFIER

Aim:-

To simulate the following circuits in proteus

(i) Halfwave rectifier

(ii) Fullwave rectifier

Apparatus required:-

Laptop with proteus software

Theory:

A rectifier is a circuit that converts alternating current (AC) into direct current (DC)

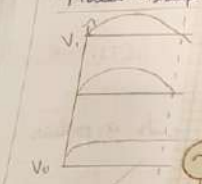
Halfwave rectifier:-

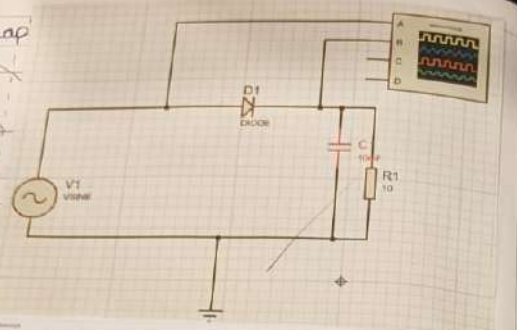
Half wave rectifier conducts only during positive half cycle. During positive half cycle the diode conducts and the output voltage is equal to the input voltage. During negative half cycle, the diode does not conduct and the output voltage is equal to zero.

Full wave rectifier:-

Full wave rectifier conducts during both positive and negative half cycles. During positive half cycle, the output voltage is equal to input voltage. During negative half cycle, the output voltage is equal to negative of input voltage. The output of

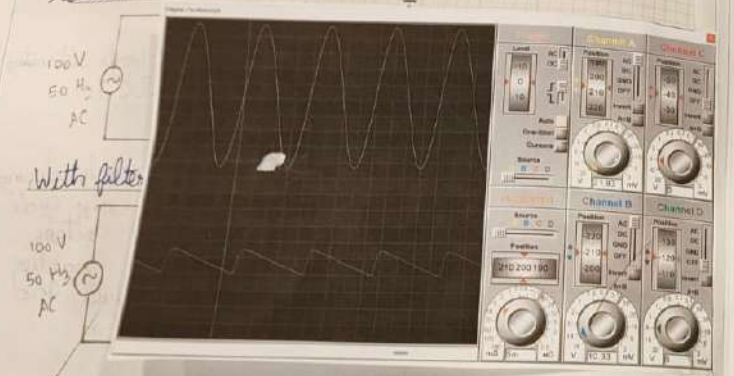
Model Graph





Circuit diagram

Fullwave rectifier



Tabulation :

Input voltage		Output voltage			
		Without filter		With filter	
V_m (V)	t (ms)	V (V)	t (ms)	V_M (V)	t (ms)
50V	20MS	50V	5MS	100V, 14V	5MS, 14MS

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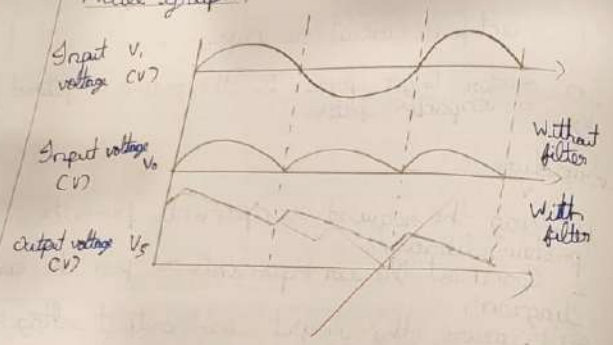
the rectifier circuit is impure DC

In order to get pure DC the output is filtered by a capacitor filter

Procedure :

1. Drag the required components from the protous library
2. Connect the components as per the circuit diagram
3. Measure the input and output voltage and note in tabulations
4. Draw the graph for input and output voltages

Model Graph:-



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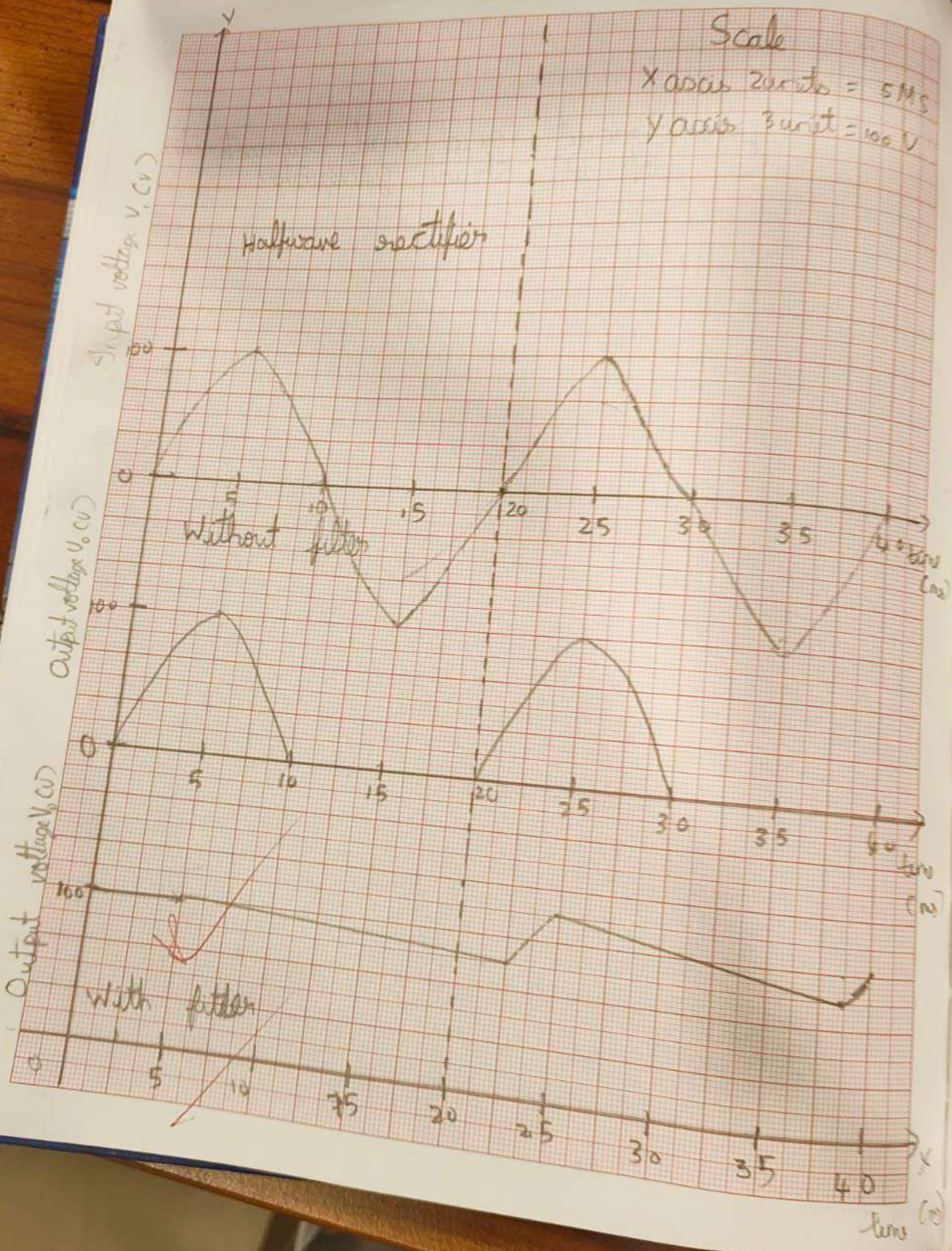
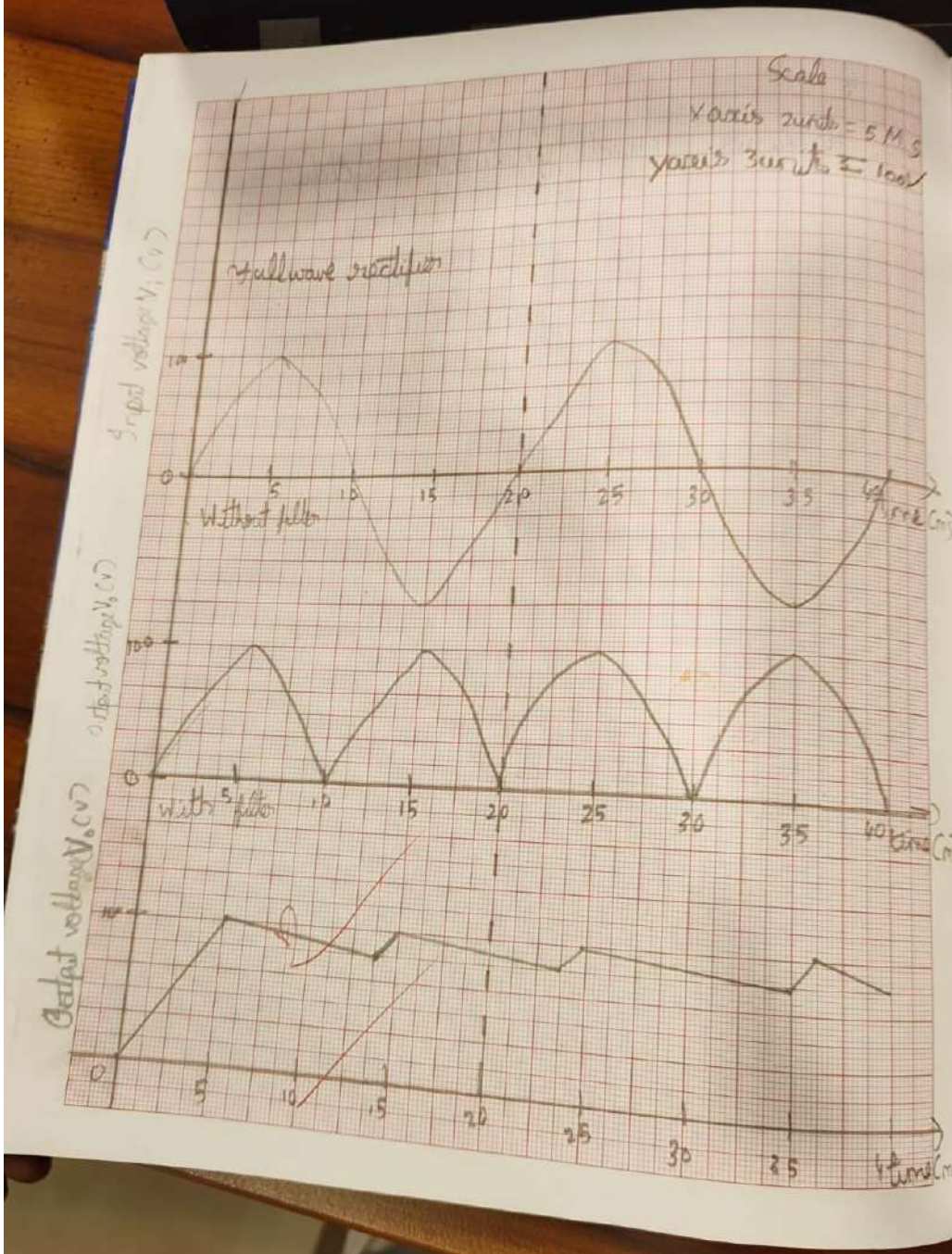
Result:-

Input and output waveforms of a half and full bridge-wave with/without filter are observed and plotted.


Scale
X axis 2 units = 5 MS
Y axis 3 unit = 100 V
Halfwave rectifier
Input voltage V_i (V)
Without filter
Output voltage V_o (V)
With filter
Output voltage V_o (V)
time (ms)
time (ms)
time (ms)
100
0
5
10
15
20
25
30
35
40


Scale
X axis 2units = 5 M.S
Y axis 3units = 100V
Halfwave rectifier
Input voltage V_i (V)
100
0
5
10
15
20
25
30
35
40
time (ms)
Without filter
output voltage V_o (V)
100
0
5
10
15
20
25
30
35
40
time (ms)
with filter
Output voltage V_o (V)
0
5
10
15
20
25
30
35
time (ms)